TECHNICAL SPECIFICATION FOR THE DEVELOPMENT OF THE SOFTWARE PRODUCT

“Galactic GeoStrategist”

1. Introduction

The purpose of the development is to create the Galactic GeoStrategist software product — a system for simulating cosmic processes with interactive geolocation-based control. The software is intended for strategic, operational, and tactical control of objects in a galactic environment and for visualization of events at various levels of detail.

2. Basis for Development

The basis for development includes the developer’s initiative, the need to create an innovative system for strategic control of space objects, and the absence of comparable solutions with camera-position-based interaction.

3. Purpose of Development

The software product is intended for visualization of cosmic processes, management of virtual objects, scenario modeling, situation analysis, and interactive strategic simulations.

4. Software Requirements

The software shall meet all functional, technical, ergonomic, and operational requirements specified in the subsections of this section.

4.1 Functional Requirements

4.1.1 Multi-level view system:
— tactical level for ship control;
— operational level for star system management;
— strategic level for diplomacy and economy.

4.1.2 Context-based control depending on camera zoom level, selected object, and current situation.

4.1.3 Supported subsystems:
— economic subsystem;
— tactical combat subsystem;
— diplomatic subsystem.

4.1.4 Support for dynamic events and scenario-based transitions.

4.2 Reliability Requirements

The software shall ensure correct operation with up to 10,000 active objects, resilience to user input errors, and stable saving and loading of sessions.

4.3 Operating Conditions

Operating system: Windows 11. The software is intended for indoor use under standard environmental conditions. A keyboard and mouse are required.

4.4 Hardware Requirements

Minimum configuration:
— CPU: Intel Core i5-8500 / AMD Ryzen 5 2600
— RAM: 8 GB
— GPU: NVIDIA GTX 1060 / AMD RX 580
— Storage: 50 GB

Recommended configuration:
— CPU: Intel Core i5-10400 / AMD Ryzen 5 3600
— RAM: 16 GB
— GPU: NVIDIA RTX 3060 Ti / AMD RX 6700 XT
— Storage: 80 GB SSD

4.5 Information and Software Compatibility

Compatibility with Windows 11 is required. The software shall support future functional expansion and potential porting to console platforms.

4.6 Ergonomics and Aesthetic Requirements

The interface shall adapt to the zoom level. Camera controls include WASD, Q/E, right mouse button, mouse wheel, and the Space key. Transitions between zoom levels shall be smooth and take from 0.5 to 3 seconds.

4.7 Information Security Requirements

The software shall provide local data storage, offline operation, and protection against unauthorized modification of software files.

5. Documentation Requirements

The documentation package shall include this technical specification, a user manual, zoom-system architecture documentation, and developer documentation.

6. Sources for Development

The sources for development include the project source Markdown file, analytical materials on strategy systems, and

GOST 34.602-89.

7. Appendices

Appendix A — Description of the zoom system architecture.